HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Owned the end-to-end lifecycle for **production customer models**, including churn and lifetime value (CLV) prediction, ensuring performance against holdout data and live outcomes.
- Defined and documented **data structure, entities, and attribute design** for multiple client datasets, translating requirements for the Data Modeller team to implement.
- Identified and resolved critical **data gaps and quality issues** across client datasets through collaboration with client business users and internal engineering teams.
- Explained complex **model results and performance metrics** clearly to non-technical stakeholders, translating insights into actionable recommendations for campaign optimisation.
- Developed and maintained **Python scripts for data extraction and transformation**, preparing datasets for model training and analysis across diverse client data sources.
- Collaborated with business users to define key performance indicators and build **BI dashboards** using Metabase, visualising customer engagement metrics and model impact.
- Containerized model serving applications using **Docker** and deployed them to **AWS EC2 instances**, establishing basic CI/CD pipelines with Jenkins for automated testing and deployment.

SKILLS

ML & Analytics – Python, scikit-learn, pandas, NumPy, R, RStudio, statistical analysis, regression, hypothesis testing, A/B testing, time-series analysis, Machine Learning fundamentals, evaluation metrics, model validation, churn models, CLV models, propensity models.

Data Engineering & Databases – SQL, PostgreSQL, MySQL, Oracle, ClickHouse, data modelling, dimensional modelling, data warehousing, ETL, data quality, data gaps, relational databases, columnar databases.

BI & Visualisation – Power BI, Tableau, DAX, Power Query, Metabase, BI dashboards, data visualisation, business intelligence.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, CI/CD, MLOps, experiment tracking, model registries.

Programming & Scripting – Python, Java, Spring Boot, REST APIs, shell scripting.

PROJECTS

Customer Churn Prediction Model for SBG Clients

- Engineered a Python-based feature pipeline to extract behavioural and transactional data for **predicting customer churn** from over 1M customer interaction records.
- Trained a **gradient-boosted classification model** (XGBoost) on the derived features, achieving an **AUC of 0.84** on a held-out validation set and identifying key drivers of churn.
- Developed a **FastAPI service** for real-time model inference, integrated with client systems via REST APIs, and documented assumptions for regulatory compliance.

Customer Lifetime Value (CLV) Dashboard and Reporting

- Developed a **data model in Power BI** to consolidate customer transaction history, engagement data, and demographic information for CLV calculation.
- Created interactive **Power BI dashboards** visualising CLV segments, predicted future value, and customer cohorts, enabling marketing teams to tailor engagement strategies.
- Wrote complex **DAX measures** to accurately calculate CLV across different time horizons and customer segments, enhancing business understanding of customer value.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)